

Cheaper Claude Sonnet Alternatives for Japanese-Learning Apps (July 2026)

Executive summary

For a low-volume Japanese-learning chat app that needs natural Japanese (including slang and keigo) plus reliable JSON/tool outputs, the best current cost-optimized options are:

- **OpenAI gpt-5.4-nano** – overall best fit: strong general capabilities, very low price (0.20/1.25 USD per 1M input/output tokens), robust structured outputs, mature Node SDK, and good practical Japanese reports, though officially positioned as a “sub-agent” model rather than a flagship writer.^{[1][2][3][4][^5]}
- **Google gemini-2.5-flash-lite** – cheapest mainstream option (0.10/0.40 USD per 1M tokens), with solid multilingual ability, official structured outputs, and good Japanese for everyday tasks, though quality and safety tuning are generally perceived as a step below OpenAI and Anthropic for nuanced Japanese tone.^{[6][7][^8]}
- **Groq llama-3.1-8b-instant** – extremely cheap and fast hosted open-weights model (0.05/1.00 USD per 1M tokens on paid tier, plus a generous free tier), with OpenAI-compatible APIs and JSON mode, but Japanese is weaker than top proprietary models and structured-output guarantees are less reliable in practice.^{[9][10][11][12]}

If you want a clear single replacement for Claude Sonnet with a strong balance of Japanese quality, JSON guarantees, and cost, **OpenAI gpt-5.4-nano is the top recommendation**; you can pair it with a higher-end model later for rare “hard mode” explanations if needed.^{[13][2][^14]}

Baseline: Claude Sonnet 4.6 pricing and capabilities

Anthropic positions Claude Sonnet 4.6 as its mid-tier, near-flagship model, with strong coding, reasoning, and agentic behavior at materially lower cost than Opus. Current public pricing for Sonnet 4.6 is around 3 USD per 1M input tokens and 15 USD per 1M output tokens, based on multiple 2026 pricing breakdowns. It supports tool calling and JSON-like structured output, but does not offer the strict JSON Schema-backed structured-output feature that OpenAI and Groq now provide.^{[15][16][17][18][19][11]}

Japanese-language evaluations consistently place Claude 4-series models (Opus and Sonnet) in the **upper tier** for nuance and politeness control, including keigo and tone-sensitive business Japanese. Qualitative reviews note that Sonnet 4.6’s Japanese has moved closer to Opus in naturalness and “non-AI-like” phrasing, especially in business writing and customer-facing copy. As a baseline, your replacement should offer at least “good enough” Japanese for email-level text, even if it falls short of Sonnet’s very high ceiling.^{[20][21]}

Option 1 – OpenAI GPT-5.4 nano (gpt-5.4-nano)

Pricing, rate limits, and access

OpenAI's GPT-5.4 nano is marketed as its **cheapest GPT-5.4-class model**, designed for high-volume, cost-sensitive tasks such as classification, extraction, ranking, and “sub-agent” workloads. Official docs and community posts agree on current pricing of **0.20 USD per 1M input tokens and 1.25 USD per 1M output tokens**, with 400k-token context and up to 128k output tokens. This is roughly an order of magnitude cheaper on input and more than 10× cheaper on output than Claude Sonnet 4.6.^{[17][22][18][2][3][23][^4]}

OpenAI uses a tiered rate-limit system; nano shares the general GPT-5.x tiers (RPM/TPM increasing with monthly spend), and is available via the standard API along with ChatGPT-backed routing in some products. For your personal use (tens to hundreds of calls per day), Tier 1 limits (on the order of 1,000 RPM and 100k TPM for small models) are ample.^{[24][25][26][27]}

Structured outputs and Node.js support

OpenAI offers **Structured Outputs**: a JSON-Schema-based mode that guarantees model outputs conform to the provided schema, superseding older JSON mode. This is explicitly designed to “ensure the model will always generate responses that adhere to your supplied JSON schema,” and the feature is supported in the Responses API that you can drive from Node.js. Community examples demonstrate combining Structured Outputs with Zod/Pydantic schemas to get fully type-safe responses, and these are widely reported as significantly more reliable than prompt-only JSON patterns.^{[19][28][29][5][30][31][^32]}

The **official openai Node/TypeScript SDK** supports the Responses API and Structured Outputs, and can be configured to use `response_format: { type: "json_schema", ... }` with your schema definition. Migrating from Anthropic's function-calling style to OpenAI's schema-driven Responses API is mostly a matter of reshaping your JSON tools definition.^{[5][33]}

Japanese quality and suitability for your use cases

OpenAI's documentation and Japanese-language media emphasize that **GPT-5.4 (flagship) has very strong Japanese, including slang and complex keigo**, and that mini/nano inherit much of this capability in smaller, faster packages. A Japanese-language benchmarking article reports GPT-5.4 mini and nano performing strongly on a custom suite of 24 tasks including Japanese reasoning and language tests; mini scored 92.9% (S rank) and nano 79.2% (A rank), suggesting nano is competitive with strong local Japanese-tuned models while costing around 1 JPY per 24 queries.^{[34][35][36][37][^38]}

While OpenAI does not publish a dedicated Japanese MT-Bench score for nano, broader medical-education and Japanese exam benchmarks show GPT-5-series models (especially GPT-5 and Gemini-2.5 Pro) at the top tier, far ahead of older GPT-4-generation systems. Practical Japanese-language write-ups describe GPT-5.4 as capable of **flexible switching between youth slang, casual speech, and highly formal keigo**, with more “human-like” tone than earlier GPT-4-generation models.^{[39][40][37][41]}

For your specific tasks:

- **In-character Japanese replies with correct register** – GPT-5.4 nano inherits its training from GPT-5.4, and community tests show it can handle Japanese email and chat with nuanced politeness, though slightly less polished than full GPT-5.4 or Claude Sonnet on high-stakes business writing.^{[14][37][^13]}
- **Self-highlighting “new” vocabulary/grammar for N3-level learners** – Nano is positioned by OpenAI for classification/extraction workloads and performs well on Japanese coding/logic tasks; extracting token-level vocabulary or grammar points and tagging difficulty is squarely in its target use case.^{[42][2][^13]}
- **Explaining function/nuance instead of literal translation** – GPT-5.4’s Japanese-focused blog posts and product demos emphasize its ability to capture the “air” or social nuance of Japanese emails and chat rather than literal word-for-word translation, and this behavior largely carries over to mini/nano with slightly less depth.^{[37][38]}
- **Repairing broken, English-shaped Japanese** – Both flagship GPT-5.4 and mini/nano are used extensively in Japanese coding and language-learning tools, where they show strong performance at inferring intended meaning from malformed Japanese and proposing native-like alternatives.^{[38][13]}

Pros and gotchas

Pros

- **Very low cost:** 0.20/1.25 USD per 1M tokens is dramatically cheaper than Claude Sonnet (3/15) and still cheaper than Gemini 2.5 Flash (0.30/2.50).^{[25][2][17][6]}
- **High-quality Japanese:** Benefiting from GPT-5.4’s strong Japanese performance, especially on nuanced tone, while being “good enough” for casual and business communication in a learning context.^{[40][39][^37]}
- **Strong structured-output guarantees** with JSON Schema support and widespread adoption patterns in Node.js ecosystems.^{[31][19][^5]}
- **Mature JS/TS SDK and ecosystem**, with many examples and community tools designed around OpenAI’s structured outputs.^{[33][32]}

Gotchas

- Official positioning frames nano as a **sub-agent / helper model**, not as the primary creative writer; for very subtle narrative tone, mini or full GPT-5.4 may still outperform it.^{[2][4]}
- OpenAI applies fairly strict content and safety filters, which can sometimes over-sanitize edgy slang or very informal internet speech compared to open-weights models.^{[26][13]}
- As with all LLMs, fully “guaranteed” JSON still benefits from a validation + retry layer; even Structured Outputs can err on very complex schemas or extremely long content.^{[43][5]}

Option 2 – Google Gemini 2.5 Flash-Lite (gemini-2.5-flash-lite)

Pricing, access, and rate limits

Google describes Gemini 2.5 Flash-Lite as its **fastest and lowest-cost 2.5-family model**, optimized for high-throughput, lightweight tasks such as large-scale classification, extraction, and summarization. The official Google developer blog lists pricing at **0.10 USD per 1M input tokens and 0.40 USD per 1M output tokens**, making it the cheapest mainstream proprietary model in its class. It supports a 1,048,576-token input context and up to 65,536 output tokens, similar to 2.5 Flash.^{[44][7][8][6]}

Gemini APIs are available via **Google AI Studio** and **Vertex AI**, with a free tier that allows limited daily usage; however, several guides note that once billing is enabled on a project, free-tier behavior changes and all token usage becomes billable. For personal use at 20–200 calls/day, both the free tier (if carefully managed) and the base paid tier are sufficient.^{[45][46][^47]}

Structured outputs and Node.js support

Gemini 2.5 Flash-Lite supports **function calling and structured outputs**, including using JSON Schema-like definitions to have the model return structured JSON objects, and integrates with built-in tools like web search and code execution when needed. The model documentation explicitly lists “関数呼び出し” (**function calling**) and “構造化出力” (**structured outputs**) among its supported features.^{[8][6]}

For Node.js/TypeScript, Google now provides the `@google/genai` **SDK** as the official, production-ready client for the Gemini API, replacing the older `@google/generative-ai` package. The SDK supports interactions with 2.5 Flash and Flash-Lite, including function calling and structured outputs, and is documented with JS/TS examples.^{[48][49][^50]}

Japanese quality and suitability

Gemini 2.5 models are generally strong multilingual models; a 2025 paper evaluating Japanese specialty-exam performance found **Gemini-2.5 Pro** slightly behind GPT-5 but ahead of earlier Gemini models on Japanese medical questions. Flash and Flash-Lite are distilled/optimized siblings, tuned more for throughput and cost than peak quality, but they inherit much of the core multilingual competency.^{[7][39][^6]}

Google’s own announcements highlight Flash-Lite as particularly well-suited for translation and classification tasks where latency and cost are key, noting that it offers **improved quality over earlier 2.0 Flash-Lite while remaining extremely fast and cheap**. Japanese dev blogs using Gemini 2.5 Flash for TTS and narration describe its Japanese speech and text handling as natural and context-aware, with good handling of prosody and tone.^{[51][52][6][7]}

For your use cases:

- **In-character Japanese replies** – Gemini 2.5 Flash (non-Lite) is often regarded as good but slightly behind the very best models (Claude 4.x, GPT-5.x) in nuanced writing and creative

prose in Japanese; Flash-Lite tends to be another small notch down but still “acceptable” for everyday chat and informal email.^{[53][54][^55]}

- **Slang and casual speech** – Community feedback indicates Gemini is comfortable with casual Japanese but can be more conservative and safety-filtered on edgy slang compared to some open-weights models, and sometimes mixes kanji into English outputs by mistake, although this is more of an English-chat issue.^{[56][54]}
- **Business keigo** – Gemini models, including Flash-Lite, can produce polite business Japanese, but comparative reviews usually rate GPT-5-series and Claude higher for very delicate corporate communication and keigo nuance.^{[57][39][^40]}

Pros and gotchas

Pros

- **Cheapest mainstream proprietary model:** 0.10/0.40 USD per 1M tokens is half of GPT-5.4 nano input cost and roughly one third of its output cost.^{[2][7]}
- Strong multilingual capability with solid Japanese, plus Google’s ecosystem tools (Maps grounding, URL context, etc.) if you ever expand the app.^{[6][8]}
- Official, GA JS/TS SDK (@google/genai) with good documentation and examples.^{[50][48]}

Gotchas

- **Free tier complexity:** enabling billing can silently disable free tier for that project, so careful project separation is needed if you rely on free usage.^{[58][45]}
- Safety filters can be quite aggressive, sometimes over-cleaning content or refusing borderline topics, which can interfere with very colloquial or edgy internet slang practice.^{[59][54]}
- Quality is good but generally a tier below GPT-5.4 and Claude Sonnet for nuanced Japanese writing and keigo, based on user reports and comparative tests.^{[39][57][^40]}

Option 3 – Groq Llama 3.1 8B Instant (llama-3.1-8b-instant via Groq)

Pricing, free tier, and access

Groq hosts Llama 3.1 8B Instant, an 8B-parameter open-weights model served on Groq’s custom LPU hardware, offering extremely low latency (hundreds of tokens per second) and low cost. Current tracking and Groq docs show pricing of 0.05 USD per 1M input tokens and 1.00 USD per 1M output tokens, with a 128k-token context window.^{[60][10][^9]}

Groq is unusual in that it offers a generous free tier with no credit card, enforced purely via rate limits rather than per-token billing. Typical free-tier limits are around 30 requests per minute and 6,000 tokens per minute, with about 14,400 requests per day across models; this is more than enough for a personal app doing <=200 calls/day.^{[61][62][63][64][^65]}

Structured outputs and Node.js support

Groq's API is **OpenAI-compatible**, meaning you can point the official OpenAI client libraries at Groq's base URL (<https://api.groq.com/openai/v1>) and call models such as `llama-3.1-8b-instant` via the Responses or chat completions endpoints. Groq also documents **Structured Outputs** that “guarantee model responses strictly conform to your JSON schema” for certain models, and supports JSON mode and tool/function calling.^{[11][66][67][68][^69]}

However, a recent bug report and community discussion show that the **guarantee is not yet as strong as OpenAI's**: a developer testing Groq's structured outputs saw about **10% of requests fail** `json_validate_failed`, whereas the same schema with OpenAI `gpt-4.1-mini` achieved 100% success, suggesting constrained decoding is not fully enforced. Broader community experiences with JSON mode across providers confirm that success rates improve dramatically when combining JSON mode with schema validation and a retry loop, regardless of provider.^{[12][70][^43]}

Groq offers its own TypeScript and Python SDKs, and can also be used via generic OpenAI-compatible clients in Node.js, making integration relatively straightforward if you already target OpenAI-style APIs.^{[66][71]}

Japanese quality and suitability

Llama 3.1 is a strong multilingual model, but its **Japanese performance lags behind top proprietary models and Japanese-specialized local models** like Swallow or LLM-jp-4. Japanese-focused derivatives such as `Llama-3.1-Swallow-8B-Instruct` score around 0.64 on Japanese MT-Bench, putting them among the better open Japanese models but still below state-of-the-art proprietary systems.^{[72][73][74][75]}

Groq's hosted `llama-3.1-8b-instant` is a **general multilingual instruct model**, not specifically Japanese-tuned. While it can produce grammatical Japanese and handle basic politeness and simple explanations, user reports and benchmarks suggest it is noticeably weaker than GPT-5.4 and Claude for:

- **Fine-grained keigo and register control**, especially in sensitive business emails.^{[21][39]}
- **Internet slang and casual nuance**, where it can sound more textbook-like.^{[76][77]}
- **Deep nuance explanations** of Japanese phrases, where it may provide correct but shallow descriptions.^{[78][76]}

That said, for a hobby-scale app where absolute top-tier Japanese nuance is not critical, Llama 3.1 8B on Groq can be “good enough,” especially if you, as an advanced user, spot-check outputs and adjust prompts to compensate.

Pros and gotchas

Pros

- **Extremely cheap and often effectively free** at your scale, thanks to the free tier and low per-M pricing.^{[79][63][^9]}

- Very fast responses even at large context sizes due to Groq's LPU hardware, giving a snappy chat experience.^{[10][64]}
- OpenAI-compatible APIs and JSON mode, so migration from Claude/OpenAI style is relatively straightforward.^{[69][66]}

Gotchas

- Japanese fluency is **meaningfully below** Claude Sonnet, GPT-5.4, or Japanese-specialized models; you may see more slightly off phrasing and weaker keigo/slang control.^{[73][74][^72]}
- Structured outputs are marketed as guaranteed but empirically show **non-trivial failure rates** without additional validation and retries.^{[11][12]}
- Model and platform are focused on speed/throughput; documentation and examples for Japanese-specific use cases are thinner than for OpenAI or Google.

Other candidates (briefly)

Several other models are worth noting but are less attractive than the three above for your specific combination of **cost + Japanese quality + JSON reliability**.

OpenAI GPT-5.4 mini (gpt-5.4-mini)

GPT-5.4 mini offers higher quality than nano—approaching full GPT-5.4 performance on coding and agent benchmarks—at **0.75/4.50 USD per 1M input/output tokens**. It is clearly very strong on Japanese tasks as part of the GPT-5.4 family and would give you quality closer to Claude Sonnet, but its pricing is closer to Sonnet than to nano, so it is less attractive if cost is the primary driver.^{[80][13][25][1][^14]}

Gemini 2.5 Flash (gemini-2.5-flash)

Gemini 2.5 Flash, the non-Lite variant, is positioned as a **balanced price-performance model** at around **0.30/2.50 USD per 1M input/output tokens**, with strong agentic and reasoning abilities. It has better quality than Flash-Lite, including for Japanese, but its price is closer to GPT-5.4 nano's output price while still lagging GPT-5.4 models slightly in nuanced writing according to comparative reports.^{[81][53][44][39][^6]}

Japanese-specialized domestic models

Japanese research institutions and companies have released high-quality domestic models like LLM-jp-4 and others that beat GPT-4o and Qwen3-8B on Japanese MT-Bench, scoring around 7.5–7.8 vs GPT-4o's 7.3. These are excellent for local deployment or custom fine-tuning but are mostly available as open weights or via limited research APIs, and do not yet offer the kind of **production-grade, strongly-guaranteed JSON schema support** you rely on.^{[82][83][84][74][^76]}

Direct comparison table

At-a-glance comparison (July 2026)

Model	Input price (USD / 1M)	Output price (USD / 1M)	Context window	Structured outputs	JS/TS SDK	Japanese quality (practical)	Notable limits / gotchas
GPT-5.4 nano (gpt-5.4-nano)	0.20	1.25	400k	Strong JSON Schema Structured Outputs; high reliability	Official openai SDK	Very good; inherits GPT-5.4's strong Japanese, good slang/keigo for most tasks	Designed as sub-agent; slightly below flagship quality, OpenAI safety filters can over-sanitize
Gemini 2.5 Flash-Lite (gemini-2.5-flash-lite)	0.10	0.40	~1M	Supports function calling & structured outputs	Official @google/genai SDK	Good; adequate for casual/business Japanese, but usually rated below GPT-5.4/Claude for nuance	Free-tier/billing quirks; aggressive safety; slightly weaker keigo and tone control
Llama 3.1 8B Instant (Groq) (llama-3.1-8b-instant)	0.05	1.00	128k	JSON mode & structured outputs, but guarantees empirically weaker	OpenAI-compatible clients & Groq SDK	Fair; usable Japanese but weaker keigo and slang nuance vs top proprietary models	JSON guarantee not fully reliable; Japanese quality behind GPT-/Claude-class; fewer JP-specific resources

Sources: OpenAI, Google, Groq, and independent pricing aggregators and benchmarks as of July 2026.^{[60][9][11][10][7][8][^2]}

Recommendation and migration notes

For your **personal Japanese-learning chat/email app**, with 20–200 short calls/day, strong structured JSON requirements, and a desire to cut costs below Claude Sonnet while maintaining good Japanese quality, the evidence points to **OpenAI GPT-5.4 nano (gpt-5.4-nano)** as the best single replacement.

- It delivers **strong, flexible Japanese** (casual, slang, keigo) for everyday communication and explanations, drawing on GPT-5.4's top-tier Japanese performance while trading only a modest amount of quality for a large cost reduction.^{[40][37][^39]}
- Its **structured-output stack is mature and battle-tested**, with JSON Schema-based guarantees and rich examples in Node.js/TypeScript ecosystems, minimizing the risk of broken JSON in your tool-like responses.^{[19][5][^31]}
- Pricing of **0.20/1.25 USD per 1M tokens** is dramatically lower than Claude Sonnet 4.6's 3/15 USD, and competitive with or better than Gemini 2.5 Flash and Flash-Lite once you factor

in quality.^{[17][7][^2]}

If you want to push costs even lower and are willing to accept some quality drop and more JSON-handling work, Gemini 2.5 Flash-Lite is the next-best proprietary option, and Groq Llama 3.1 8B Instant is the cheapest/fastest hosted alternative, with the caveat that Japanese nuance and JSON reliability are weaker in both cases.^{[9][7][8][12]}

In practice, a pragmatic migration path would be:

1. **Switch core flows** (reply generation, N3-level vocabulary tagging, nuance explanation, learner-input repair) to gpt-5.4-nano with Structured Outputs.
2. Implement a **thin JSON validation + retry layer** around your existing function-calling/JSON schema, which will also keep the door open to using Groq or Gemini later.^{[32][43]}
3. For rare cases where you need extra polish (e.g., important business-keigo emails), optionally route those specific prompts to a higher-end model (GPT-5.4 or Claude Sonnet) via a manual “high-quality mode,” while keeping nano as the default for 90–95% of traffic.^{[13][57]}

This mix should give you **substantially lower spend than Claude Sonnet** for your usage level, while preserving the Japanese-learning UX you currently have.

References

1. [Introducing GPT-5.4 mini and nano — our most capable small...](#) - Pricing comparison Model In Cached Out gpt-5.4-mini \$0.75 \$0.075 \$4.50 gpt-5-mini \$0.25 \$0.025 \$2.00...
2. [GPT-5.4 nano Model | OpenAI API](#)
3. [OpenAI、「GPT-5.4 mini」と「GPT-5.4 nano」をリリース \(ZDNET Japan\)](#) - Yahoo!ニュース - OpenAIは米国時間3月17日、「GPT-5.4 mini」と「GPT-5.4 nano」をリリースした。どちらのモデルも、大量の人工知能（AI）ワークロードを高速かつ効率的に処理できるように設
4. [OpenAI、2倍以上高速の小型AI「GPT-5.4 mini/nano」提供開始 \(PC Watch\)](#) - Yahoo!ニュース - OpenAIは3月17日、新たな小型AIモデル「GPT-5.4 mini」および「GPT-5.4 nano」を提供開始した。どちらもGPT-5.4の特徴を受け継ぎつつ、高速かつ効率的な動作を実現す
5. [OpenAI Structured Outputs: JSON Schema Guide — Jsonic](#) - response_format json_schema with strict mode — 100% schema-compliant JSON from OpenAI models, with t...
6. [Gemini 2.5: 思考モデル ファミリーのアップデート](#) - Explore the latest Gemini 2.5 model updates with enhanced performance and accuracy: Gemini 2.5 Pro a...
7. [Gemini 2.5 Flash-Lite、安定版として一般提供を開始](#) - Explore Gemini 2.5 Flash-Lite, Google's stable and generally available model offering incredible spe...
8. [Gemini 2.5 Flash-Lite | Gemini API | Google AI for Developers](#) - Google の Gemini 2.5 Flash-Lite について
9. [Llama 3.1 8B Instant — API Pricing & Specs | Groq — Model Price Watch](#) - Llama 3.1 8B Instant by Groq. Input \$0.05/Mtok, Output \$1/Mtok. 128K token context. Llama 3.1 8B on ...
10. [Llama 3.1 8B - GroqDocs](#) - Model card for Llama 3.1 8B: 8B parameter model with 128K context, tool use, JSON mode, and instant ...

11. [Structured Outputs - GroqDocs - Groq Console](#) - Guarantee model responses strictly conform to your JSON schema for reliable, type-safe data structur...
12. [Guaranteed structured output is not working.\(bug report\) - Forum](#) - The documentation for Structued Output claims that models gpt-oss-120b and gpt-oss-20bsupport guaran...
13. [\[Thorough Verification\] GPT-5.4 mini & nano Released. Comparative ...](#) - On March 17, 2026 (JST), OpenAI released the new models "GPT-5.4 mini" and "GPT-5.4 nano".These are ...
14. [OpenAI、高速・低コストに特化した軽量AIモデル「GPT-5.4 mini」および「nano」を公開（ビジネス+IT）](#) - 米OpenAIは2026年3月17日、最新AIモデル「GPT-5.4」の軽量版となる「GPT-5.4 mini」と「GPT-5.4 nano」を発表した。推論速度とコスト効率を重視した設計で、A
15. [Introducing Claude Sonnet 4.6](#) - Claude Sonnet 4.6 is a full upgrade of the model's skills across coding, computer use, long-reasonin...
16. [Claude Sonnet 4.6: Complete Specifications, Pricing, API ...](#) - Claude Sonnet 4.6: Complete Specifications, Pricing, API Access & Use Cases (2026) | [Gate.AI](#)
17. [Claude Sonnet 4.6: Specs, Benchmarks & Pricing 2026](#) - Claude Fast | Sonnet 4.6 delivers Opus-level intelligence at \$3/\$15 per million tokens. 70% preferre...
18. [Claude API Pricing 2026: Opus 4.8, Sonnet 4.6, Haiku 4.5 Costs](#) - Claude API pricing May 2026: Opus 4.8 & 4.7 (\$5/\$25), Sonnet 4.6 (\$3/\$15), Haiku 4.5 (\$1/\$5) per MTo...
19. [Structured model outputs | OpenAI API](#) - Structured Outputs is a feature that ensures the model will always generate responses that adhere to...
20. [新登場！Claude Sonnet 4.6の日本語文章作成能力を深堀調査](#) - この記事はこんな人向け AIで日本語の文章を書いているが「なんかAIっぽい」から脱却したい人 Claude Sonnet 4.6の日本語文章力が実際にどう変わったか知りたい人 目的別のプロンプトテンプレ...
21. [Claude Opus 4.6の驚異的な日本語性能－ベンチマークに ...](#) - 敬語と丁寧語の使い分けはもちろん、場の空気や人間関係を踏まえた ... Claude Sonnet 4.6がDatabricksにネイティブ統合！AIエージェント構築 ...
22. [GPT 5.4 Nano by OpenAI — Pricing, Specs & API Access](#)
23. [GPT-5.4 mini 和 nano 登場](#) - GPT-5.4 mini 和 nano 是 GPT-5.4 更精簡、更快速的版本，專為程式碼編寫、工具使用、多模態推理，以及高流量 API 與子智慧體任務進行最佳化。
24. [o3-mini Model | OpenAI API](#)
25. [Pricing | OpenAI API](#) - Pricing information for the OpenAI platform.
26. [OpenAI API Billing 2026: Credits, 5 Tiers, Auto-Recharge](#) - OpenAI API billing 2026: prepaid credits, auto-recharge, 5 usage tiers, spending caps. Common surpri...
27. [OpenAI API Rate Limits: GPT-5.5, GPT Image 2 & Free Tier \(2026\)](#) - Full 2026 OpenAI rate limits by model and tier. Compare RPM, TPM, and batch limits for GPT-5.5, GPT-...
28. [Introducing Structured Outputs in the API](#) - Today we're introducing Structured Outputs in the API, a new feature designed to ensure model-genera...
29. [APIに Structured Outputsを導入 - OpenAI](#) - APIに Structured Outputs（構造化出力）を導入することにより、モデルの出力と開発者指定 JSON スキーマの一致が確保されます。
30. [OpenAI Structured Outputs Tutorial \(Parse data string to beautiful JSON\)](#) - Extracting Structured Data with OpenAI's New Feature Using Node.js

In this video, we demonstrate ho...

31. [OpenAI Structured Output Tutorial \(Responses API + JSON ...](#) - Structured output is a feature that ensures that the model will always generate responses that adher...
32. [OpenAI JSON Mode and Function Calling: Structured JSON Output ...](#) - Get structured JSON from OpenAI GPT models using JSON mode, function calling, and the Responses API....
33. [openai-node/helpers.md at f4f5fdefc3b4d2ef0fd7b4bd18af858c7a9b9eeb · openai/openai-node](#) - Official JavaScript / TypeScript library for the OpenAI API - openai/openai-node
34. [GPT-5.4 mini と nano が登場 - OpenAI](#) - GPT-5.4 mini と nano は、コーディング、ツール利用、マルチモーダル推論、高負荷の API 処理やサブエージェントのワークロード向けに最適化された、GPT-5.4 のより小型で高速なバ...
35. [93% accuracy for ¥2, 80% for ¥0.2: Testing GPT-5.4 mini/nano on 24 ...](#)
36. [¥2で93%、電気代¥0.2で80%——GPT-5.4 mini/nano を同じ ...](#)
37. [GPT-5.4の日本語能力を徹底検証！日程変更・お断りメールの自動...](#) - 導入本記事を読めば、最新のGPT-5.4が持つ「日本語の空気感」を読み取る力を理解し、これまで躊躇していた顧客対応や日程変更メールの自動化へ自信を持って踏み切れるようになります。「この言い回し、少し冷...
38. [【速報】GPT-5.4 \(Thinking\) 正式リリース！日本語対応や旧...](#) - 導入2026年3月5日、OpenAIから最新モデルとなるGPT-5.4 (Thinking) が正式にリリースされました。単なるテキスト生成の枠を超え、「自律的に思考し、計画を立てる」プロセスが実装された...
39. [Performance of large language models on the Japanese ... - PubMed](#) - Successive model generations, particularly GPT-5 and Gemini-2.5Pro, consistently achieved passing-le...
40. [Gemini Advanced \(Google AI Pro\) と ChatGPT Plus の包括的比較](#) - GPT-5.4は、日本語のスラングや若者言葉から、非常に複雑なビジネス敬語までを柔軟に使い分けることができ、その文章は「人間味がある」「寄り添い感がある」と評される ...
41. [GPT-5.4のベンチマーク・使用評価・反応に関する公開情報 ...](#)
42. [GPT-5.4 mini/nano Subagentアーキテクチャ実践ガイド - Shareuhack](#) - GPT-5.4 mini (\$0.75/\$4.50 per 1M tokens) と nano (\$0.20/\$1.25) はsubagentアーキテクチャ向けに設計されています。Planner-Executo...
43. [Structured output reliability with LLMs — 3-month production ...](#) - A developer shares production learnings from a health app using LLMs for structured JSON output. The...
44. [Gemini 2.5 Flash | Gemini API - Google AI for Developers](#) - Learn about Gemini 2.5 Flash from Google
45. [Gemini API Free Tier Limits 2026: RPM, RPD & TPM by ...](#) - The 2026 Gemini API free-tier limits by model (RPM, RPD, TPM) - plus the catch the docs bury: enabli...
46. [Gemini API Free Tier 2026: Limits, Quotas, and More](#) - Complete guide to Google's Gemini free tier in 2026. Google AI Studio limits, Vertex AI free credits...
47. [Gemini API Pricing 2026: All Tiers & Free Limits | DeployBase](#) - Gemini API pricing 2026: Pro and Flash tiers, free tier limits, batch API discounts as of March 2026...
48. [Gemini API libraries | Google AI for Developers](#) - Download and get started with the Gemini API Libraries + SDKs

49. [Getting started with the Gemini API and Web apps](#) - Learn how to use the Gemini API and the Google Gen AI SDK for JavaScript and TypeScript to prototype...
50. [Gemini API | Google AI for Developers](#) - Gemini API Docs and API Reference
51. [Google、音声生成モデル「Gemini 2.5 Flash Native Audio」をアップデート／指示に忠実、より自然なやりとりを実現。ライブ音声翻訳など自社製品でも活用](#) - 米Googleは12月12日（現地時間）、ライブ音声エージェント向けに最適化されたAIモデル「Gemini 2.5 Flash Native Audio」のアップデートを発表した。「Google AI...
52. [Gemini 2.5 Flash TTS で日本語ナレーション音声をAPI生成する方法](#) - はじめに GoogleのGemini 2.5 Flash Preview TTSを使うと、REST APIだけで自然な日本語ナレーション音声を生成できます。従来のGoogle Cloud TTSと比べ...
53. [Gemini 2.5 シリーズ完全ガイド：Pro, Flashの性能&新登場...](#) - Gemini 2.5 Pro・Flash・Flash-Liteの3モデルを音声文字起こしとSVG生成で実測比較。無料で使えるGoogle AI Studioでの利用法も解説する
54. [Gemini 2.5: さらに進化した、これまでで最も高性能なモデル](#) - Gemini 2.5 Pro は、コーディングに最適なモデルとして引き続き開発者の皆様に高く評価されており、2.5 Flash も新たなアップデートによりさらに進化しています。今回、Gemini 2....
55. [Gemini 2.5 Pro徹底レビュー | 料金・性能・ChatGPT比較【2026...](#) - クイックサマリー：ChatGPTと比べてgemini
56. [Gemini 2.5 Flash is randomly including kanji into English conversation](#) - Gemini 2.5 Flash is randomly including kanji into English conversation
57. [Claude・Geminiとの実機比較結果はいかに?! GPT-5.4の...](#) - OpenAIの最新モデルGPT-5.4の概要・価格・ベンチマークを解説。Claude Sonnet 4.6・Gemini 3.1 Proとの実機比較で日本語要約・コード生成・構造化タスクの性能差を検証...
58. [Google Gemini API free tier tightened: Pro models to become paid...](#) - Author's Note: A detailed breakdown of the major changes to the Google Gemini API free tier as of Ap...
59. [Gemini 2.5 Flash Review \(2026\) — Gemini 2.5 Flash](#) - A multimodal AI model praised for cheap, high-quality image editing, but facing criticism for coding...
60. [Groq API Pricing \(Jul 2026\): 7 models from \\$0.065/Mtok — Model Price Watch](#) - Groq API pricing (July 2026): every current model with input, output, and cached-input prices per 1M...
61. [Rate Limits - GroqDocs](#) - Understand Groq API rate limits, headers, and best practices for managing request and token quotas i...
62. [Groq Free Tier, Signup Credits, and Limits](#) - Check Groq free tier, signup credits, model limits, free API access, Mainland China accessibility, a...
63. [Groq Free Tier 2026 — Free Models, Credits & Limits | Price Per Token](#) - Groq offers a genuinely free, no-credit-card developer tier with access to every model on its ultra...
64. [Groq API Pricing 2026: Free Tier, 315 TPS, \\$0.05/M Paid...](#) - Groq API pricing 2026: free tier 30 req/min, paid from \$0.05/M. 300-1,000 TPS speed. Rate limits by ...
65. [Groq Rate Limits & Usage Caps 2026 | Tickerr](#) - Real-time status and API pricing for 90+ AI tools - ChatGPT, Claude, Gemini, Grok, Cursor and more. ...

66. [OpenAI Compatibility - GroqDocs](#) - Learn how to use OpenAI's client libraries with Groq API, including configuration, supported feature...
67. [Introduction to Tool Use - GroqDocs](#) - Learn why tools are essential for building powerful AI agents and how they enable LLMs to take actio...
68. [groq.package - github.com/nocturnium/llm- ...](#) - Package groq provides a Groq LLM client using native HTTP.
69. [Overview - GroqDocs](#) - Fast LLM inference, OpenAI-compatible. Simple to integrate, easy to scale. Start building in minutes...
70. [Structured output reliability with LLMs — 3-month production learnings](#) - Structured output reliability with LLMs — 3-month production learnings
71. [Groq.provider - AI SDK](#)
72. [Llama 3.1 Swallow](#) - The Japanese MT-Bench average score for Llama 3.1 Swallow 8B Instruct v0.3 is 0.6424, placing it amo...
73. [Introducing Llama 3.1: Our most capable models to date](#) - we're introducing upgraded versions of the 8B and 70B models. These are multilingual and have a sign...
74. [国産LLM「LLM-jp-4」が日本語MT-BenchでGPT-4oを上回った ...](#) - 2026年4月3日、国立情報学研究所 (NII) が国産LLM「LLM-jp-4」を公開しました。日本語MT-Benchで GPT-4o のスコアを上回ったという発表は、国内外で大きな注目を集めています。 ...
75. [tokyotech-llm/Llama-3.1-Swallow-8B-Instruct-v0.5 at d4ba6fb](#) - We're on a journey to advance and democratize artificial intelligence through open source and open s...
76. [日本語 LLM ベンチマークとリーダーボード一覧](#) - Nejumi 4、Swallow v2、Open Japanese LLM など6つのリーダーボードと20以上のベンチマークを横断的にまとめた。日本産モデル10種のスコア比較、中国モデルの漢字アドバン...
77. [RTX 5080で動かす日本語ローカルLLM比較](#) - RTX 5080とは、NVIDIA Blackwell世代のハイエンドGPU (VRAM 16GB) である。ローカルで自然な日本語チャットを動かしたいなら、選択肢は2つに集約されます。日本語特化の8B...
78. [AI Agent API Benchmark Report — February 2026 | washinmura](#) - 30+ API real benchmarks: 15 LLM rankings, search engine comparison, translation evaluation, 24h stab...
79. [Groq Pricing 2026: Per-Model Cost, Free Tier, Speed ...](#) - Groq pricing in 2026: Llama 3.1 8B \$0.05/M, Llama 70B \$0.59/M, Kimi K2 \$1/M. Free tier limits, Batch...
80. [How Much is GPT-5.4 Mini & Nano? \(2026 Costs\)](#) - Discover the exact 2026 API costs for GPT-5.4 Mini (\$0.75) and Nano (\$0.20). Learn the hidden fees, ...
81. [Gemini API Pricing: Current Flash, Flash-Lite, and Pro Rates](#) - Current Gemini API prices per 1M tokens: Gemini 3.5 Flash \$1.50/\$9, Gemini 3.1 Pro \$2/\$12, Gemini 3 ...
82. [NII develops new domestic LL trained on a high-quality corpus of 12 ...](#) - NII develops new domestic LL trained on a high-quality corpus of 12 trillion tokens
83. [日本特有の知識に特化した大規模言語モデルの学習および評価](#)
84. [Release of New Japanese LLMs](#) - As Japan's only general academic research institution seeking to create future value in the new disc...

